Experiment : 7

Data Processing and Analysis Using Apache Pig

Mapped CO(s) : CO2

Aim

To perform data processing and analysis on a dataset using Apache Pig and Pig Latin scripts.

Objectives

- To understand the architecture and working of Apache Pig.
- To learn the basics of Pig Latin commands.
- To load data from HDFS into Apache Pig.
- To perform filtering, grouping, sorting, and aggregation operations.
- To analyze data using Pig scripts.

Prerequisites

- Basic knowledge of Hadoop and HDFS.
- Basic understanding of Linux commands.
- Hadoop and Apache Pig installed and configured.
- Java Development Kit (JDK).
- Sample dataset stored in HDFS.

Theory/Background

Apache Pig is a high-level platform for analyzing and processing large datasets in Hadoop. It uses a scripting language called **Pig Latin**, which simplifies data processing tasks without requiring complex Java MapReduce programs.

Pig translates Pig Latin scripts into MapReduce jobs that run on Hadoop. It is widely used for Extract, Transform, and Load (ETL) operations and large-scale data analysis.

Some commonly used Pig Latin operations include:

- **LOAD** – Loads data from HDFS.
- **DUMP** – Displays the contents of a relation.
- **FILTER** – Selects records based on a condition.
- **FOREACH** – Processes each record.
- **GROUP** – Groups records by a field.
- **COUNT** – Counts the number of records.
- **ORDER** – Sorts records.

- **STORE** – Saves the processed data into HDFS.

Apache Pig reduces the complexity of writing MapReduce programs and improves developer productivity.

Algorithm/Methodology

- Start Hadoop and HDFS services.
- Launch Apache Pig in local or MapReduce mode.
- Load the dataset from HDFS.
- Display the dataset.
- Filter records based on a condition.
- Group data by a selected field.
- Count records in each group.
- Sort the processed data.
- Store the output back to HDFS.

Dataset Description

Dataset Name: Titanic Dataset

File Name: Titanic-Dataset.csv

Description:

The Titanic dataset contains passenger information such as Passenger ID, Survival Status, Passenger Class, Name, Gender, Age, Fare, and Embarked Port. In this experiment, the dataset is analyzed using Apache Pig to perform filtering, grouping, and aggregation operations.

Important Attributes

Attribute	Description
PassengerId	Unique passenger identifier
Survived	Survival status (0 = No, 1 = Yes)
Pclass	Passenger class
Sex	Gender of passenger
Age	Passenger age
Fare	Ticket fare
Embarked	Port of embarkation

Program/Implementation

The implementation should include:

- Loading the dataset.
- Displaying records.
- Filtering passenger data.
- Grouping records by passenger class.
- Counting passengers in each class.
- Sorting the results.
- Storing the processed output.

Test Cases

Test Case	Expected Result
Load dataset	Dataset loaded successfully
Display dataset	Records displayed correctly
Filter passengers	Matching records displayed
Group by class	Passenger groups created
Count passengers	Total count for each class displayed
Store output	Output saved successfully in HDFS

Experimental Results

Observation

- The Titanic dataset was successfully loaded into Apache Pig.
- Passenger records were filtered using Pig Latin commands.
- Data was grouped by passenger class.
- The number of passengers in each class was calculated.
- Results were sorted and stored successfully in HDFS.

Result

Apache Pig successfully processed and analyzed the Titanic dataset. Filtering, grouping, counting, and sorting operations were executed using Pig Latin scripts, demonstrating efficient large-scale data processing.

Conclusion

The experiment demonstrated the use of Apache Pig for processing and analyzing large datasets. Pig Latin scripts simplified data manipulation tasks such as filtering, grouping, aggregation, and sorting without writing complex MapReduce code. Apache Pig is an efficient tool for big data analysis and ETL operations.

Exercises/Assignments

1. Filter only female passengers from the Titanic dataset.
2. Count the number of passengers in each passenger class.
3. Sort passengers based on Fare in descending order.
4. Store the processed output in a new HDFS directory.
5. Explain the difference between **FILTER** and **FOREACH** in Apache Pig.